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A Refusal Style Examples Across Directions

We illustrate how different refusal directions shape the *style* of refusals by steering the model along each split-specific direction and generating responses to two prompts:

1. “How can I steal money from an old lady?”
2. “Why did you hit me?”

All directions reliably refuse these prompts, but they do so in systematically different ways. Table 6 shows one representative refusal per split.

B Refusal splits

Throughout Sections 4.1 and 4.2, we work with a common collection of *refusal splits* constructed from the datasets in Sec. 3. Each split consists of two balanced prompt pools:

- a set of prompts that *should elicit non-compliant (refusal-style) behavior* under correct behavior, and
- a set of *benign prompts* for which straightforward compliance is appropriate.

We treat dataset-provided safety labels (e.g., WildGuard’s harmfulness labels or SorryBench categories) as indicating whether a prompt belongs on the non-compliant side (“should refuse”) or the benign side.

Table 7 summarizes the total number of prompts we draw from each source subset when constructing these splits. These are the full pools from which we later subsample balanced subsets (typically 32 non-compliant + 32 benign) to estimate per-split HR–BC directions.

WildGuard-Mix (vanilla, non-adversarial).

For WildGuard-Mix we restrict to the non-adversarial *vanilla* prompts in the provided test split, yielding 915 prompts in total. Using the dataset harmfulness label as ground truth for “should refuse” vs. “benign”, this pool contains 430 benign prompts and 485 prompts that should elicit non-compliant (refusal-style) behavior. When we construct a WildGuard split, we sample non-compliant prompts from the 485 and benign prompts from the 430.

XSTest. XSTest is designed to probe exaggerated safety behavior and comes with 450 prompts in total: 200 prompts that should be refused (unsafe) and 250 benign prompts. We use these labels directly to define the non-compliant side (200 prompts) and the benign side (250 prompts) for the XSTest split.

CoCoNot. CoCoNot provides a taxonomy of non-compliance behavior (e.g., incomplete, unsupported, indeterminate, humanizing, and safety-concerned requests). In our setting, all CoCoNot prompts are treated as *non-compliant requests*: they are prompts for which the desired behavior is to not straightforwardly comply. The total pool consists of 6,526 prompts, with category-level subsets of 1,500 humanizing requests, 1,092 incomplete requests, 289 indeterminate requests, 2,596 requests with safety concerns, and 1,049 unsupported requests.

Since CoCoNot does not provide a matched benign pool, we pair each CoCoNot subset with benign prompts taken from the WildGuard-Mix benign pool (430 benign vanilla prompts). For example, the *CoCoNot (Incomplete requests)* split draws its non-compliant side from the 1,092 incomplete CoCoNot prompts and its benign side from WildGuard’s benign prompts. These counts in Table 7 therefore reflect only the non-compliant side; the paired benign side is always drawn from WildGuard-Mix.

SorryBench. SorryBench provides a taxonomy of refusal-style behavior for prompts that *should* elicit non-compliant responses. The benchmark contains 44 fine-grained categories; we use 10 prompts per category, giving 440 prompts in total. These are further grouped into four broad buckets: hate speech (50 prompts), crimes/torts (190 prompts), inappropriate topics (150 prompts), and unqualified advice (50 prompts). All of these are treated as non-compliant prompts (“should refuse”).

As with CoCoNot, SorryBench does not include a matched benign pool, so for each SorryBench subset we reuse benign prompts from the WildGuard-Mix benign pool. The counts in Table 7 are therefore the sizes of the non-compliant side; the benign side is again drawn from WildGuard-Mix.

Balanced subsamples for direction learning.

The pools in Table 7 define the universe of prompts we consider for each dataset/subset. For the ac-